Sahaj Bhadja

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Summary

Software Engineer with an MS in Computer Science and 2+ years in **Python** and **C++**, building **automated test** systems, data analysis pipelines, and simulation-based validation, contributing to team codebases with engineers across disciplines.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built **Python** data pipelines with incremental loads, validation, and reconciliation checks across SQL databases and cloud storage, sustaining 99%+ reliability over 5K+ monthly production jobs through CI/CD releases.
- Built fault-tolerant concurrent job processing with explicit state transitions and failure isolation, handling 1K+ jobs at 3x throughput, and tuned queries to cut **dashboard** latency from 1.2s to 350ms.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Designed a **multithreaded** batch-processing engine in Python with a thread pool and fault-tolerant execution, processing 1K+ files per run while keeping services stable for 150+ instructors at peak load.
- Tracked every model run with per-call usage and cost logs in MySQL and **measured** grading consistency at 90%+ across 10K+ submissions, cutting instructor feedback turnaround 60% in production use.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built Python pipelines processing 100K+ vehicle trajectory logs sampled at 10Hz into aligned **measurement** features for training and evaluating a PyTorch planner, improving long-horizon planning 18% in simulation.
- Ran controlled **test** comparisons of a new encoder against three baselines in closed-loop simulation with Weights & Biases tracking, documenting failure modes and improving planning stability 15%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built and maintained an **automated test system** that compiled builds nightly, installed them on test machines across three operating systems, ran 20+ suites, and generated CSV reports and failure logs.
- Contributed object-oriented Java and Python modules with Singleton and Factory patterns to a **large shared codebase**, improving response times 30% and cutting technical debt 20% with Microsoft stakeholders.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst – Research Fellowship *India*

- Engineered 21 behavioral features over a 650K+ URL dataset with statistical **data analysis**, and trained Gaussian Naive Bayes and Random Forest classifiers to detect phishing and malware patterns.
- Compared classical models against RNN, LSTM, Bi-LSTM, GRU, and CNN architectures, achieving **96%+ detection accuracy** and documenting trends in phishing and malware behavior across the full dataset.

Skills

Languages: Python, C++, Java, SQL, JavaScript, Bash, Shell
Test & Analysis: Test Automation, Unit Testing, Integration Testing, PyTest, Mockito, Data Analysis, Feature Engineering
Systems: Linux, Windows, Concurrency, Multithreading, Operating Systems, Data Structures, Algorithms, Simulation
Data & ML: PyTorch, TensorFlow, Data Pipelines, PostgreSQL, MySQL, Tableau, Weights & Biases
Tools: Git, Jenkins, CI/CD, Docker, Debugging, Object-Oriented Design, Design Patterns, Agile, Jira, AWS, Selenium

Projects

Instrument Test Automation Framework | *Python, PyVISA, SCPI, pytest, pandas*

- Designed a **Python** framework that controls signal generators and **oscilloscopes** over SCPI, runs parameter sweeps on a device under test, captures waveforms, and logs pass or fail results per test.

Waveform Analysis Toolkit | *C++17, Python, NumPy, matplotlib, GoogleTest*

- Designed a **C++** and Python toolkit that filters captured signals, measures amplitude, noise, and timing jitter, and plots **trend reports** that flag drift across repeated test runs.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2, OS Scheduling*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel queue **scheduling** in C++ with Gantt visualization, testing 1K+ process combinations and cutting average turnaround time 25% across test scenarios.